

STOCKHOLM OBSERVATORIET, Sweden

WMO: 024840

Lat: 59.3415N

Lon: 18.0546E

Elev: 44

StdP: 100.80

Time Zone: 1.00 (EUC)

Period: 01-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-12.6	-9.9	-14.7	1.1	-10.8	-12.7	1.3	-8.1	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	7.7	27.3	18.3	25.5	17.4	23.8	16.6	19.9	24.2	18.8	23.0	17.9	22.0	N/A	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.5	13.4	21.3	17.4	12.5	20.4	16.2	11.6	19.6	57.1	24.6	53.6	23.0	50.5	22.0	23.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature								
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years		
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
N/A	N/A	N/A	DB	-13.8	29.9	3.6	1.6	-16.3	31.0	-18.4	32.0	-20.4	32.9	-23.1	34.0
			WB	-14.1	20.9	3.5	1.5	-16.6	21.9	-18.7	22.8	-20.7	23.6	-23.3	24.7

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.0	-1.4	-1.1	1.6	6.6	11.8	15.8	19.1	17.9	13.3	7.6	3.6	0.6		
	DBStd	7.98	4.31	4.06	4.03	3.50	3.57	3.15	2.97	2.62	3.02	3.28	3.41	4.22		
	HDD10.0	1635	354	311	261	112	21	1	0	0	8	87	191	291		
	HDD18.3	3856	613	544	519	351	203	88	26	40	151	332	441	549		
	CDD10.0	911	0	0	0	11	78	176	282	244	108	13	0	0		
	CDD18.3	89	0	0	0	0	2	12	49	25	1	0	0	0		
	CDH23.3	466	0	0	0	0	14	72	287	91	1	0	0	0		
	CDH26.7	78	0	0	0	0	1	7	59	11	0	0	0	0		
(14) Wind	WSAvg	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(15) Precipitation	PrecAvg	550	39	27	26	31	32	50	72	64	56	50	50	47		
	PrecMax	800	74	72	69	87	80	113	192	162	171	137	104	117		
	PrecMin	376	6	4	1	6	7	3	17	3	13	9	16	9		
	PrecStd	94	17	15	14	18	18	26	44	36	36	31	23	25		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.6	8.2	13.8	19.5	25.2	27.8	30.5	28.2	22.9	15.8	11.6	10.7		
		MCWB	5.9	5.8	7.7	11.0	14.9	17.6	18.8	19.2	16.4	12.8	10.5	8.9		
	2%	DB	6.1	6.3	10.8	16.8	22.6	25.4	28.4	25.7	20.7	14.3	10.2	8.3		
		MCWB	4.9	4.2	6.1	9.8	13.9	16.2	18.6	18.1	15.2	12.1	9.1	6.9		
	5%	DB	5.0	5.1	8.9	14.6	20.2	23.5	26.5	23.8	19.1	13.1	9.2	6.9		
		MCWB	4.0	3.6	5.2	8.6	13.0	15.3	18.2	17.4	14.8	11.5	8.2	5.6		
	10%	DB	3.9	3.9	7.0	12.4	18.0	21.6	24.6	22.3	17.6	12.1	8.1	5.7		
		MCWB	3.0	2.8	4.0	7.2	11.8	14.5	17.3	16.8	13.9	10.7	7.2	4.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.5	6.7	8.2	12.0	16.5	18.7	21.5	21.5	17.2	13.8	10.8	9.0		
		MCDB	7.3	7.9	12.6	17.6	23.0	25.2	26.2	25.5	21.1	15.0	11.3	10.4		
	2%	WB	5.2	4.7	6.6	10.3	15.0	17.3	20.2	19.8	16.1	12.6	9.4	7.1		
		MCDB	5.9	5.8	10.0	15.9	20.3	23.2	25.0	23.0	19.2	13.7	10.1	8.0		
	5%	WB	4.1	3.7	5.5	9.0	13.6	16.2	19.2	18.6	15.3	11.8	8.3	5.8		
		MCDB	4.9	4.8	8.3	13.9	18.6	21.5	23.9	21.9	18.0	13.0	9.0	6.6		
	10%	WB	3.1	2.8	4.5	7.7	12.4	15.2	18.3	17.7	14.6	10.9	7.3	4.7		
		MCDB	3.8	3.8	6.7	11.6	17.1	20.1	22.8	20.9	17.0	11.9	8.0	5.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	3.5	3.8	5.4	7.5	8.3	7.9	7.7	7.0	5.9	4.3	3.2	3.4		
		MCDBR	4.1	4.4	8.0	10.6	11.7	11.1	10.4	9.1	7.7	4.9	4.0	4.0		
	5% WB	MCWBR	4.0	3.4	4.8	5.6	5.7	4.7	3.5	3.4	3.5	3.5	3.8	3.8		
		MCWBR	4.1	4.0	7.4	9.6	10.4	9.7	8.9	7.5	6.8	4.8	3.8	3.8		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.321	0.339	0.375	0.372	0.362	0.390	0.381	0.348	0.336	0.309	0.267		
		taud	2.219	2.287	2.355	2.324	2.397	2.457	2.400	2.420	2.497	2.417	2.254	2.140		
	Ebn at Noon		516	687	797	826	858	872	834	814	788	669	473	384		
		Edh at Noon	44	69	87	106	106	101	105	96	76	61	43	32		
(44) All-Sky Solar Radiation	RadAvg		0.29	0.87	2.38	4.01	5.32	5.91	5.47	4.32	2.79	1.22	0.39	0.20		
	RadStd		0.04	0.14	0.26	0.38	0.45	0.34	0.51	0.38	0.30	0.20	0.09	0.02		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (51 neighbors)	+0.37	N/A	N/A	N/A	N/A	N/A	-116	-145	N/A	N/A

Nomenclature: See separate page